An Effiecient Deep Learning Pipeline for Quadrant Detection, Tooth Enumeration, and Disease Detection from Panoramic Radiographs

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Abstract

Timely and accurate diagnosis of dental pathologies from panoramic radiographs is a critical yet resource intensive task in clinical dentistry. This paper presents a pipeline that automates quadrant detection, tooth enumeration, and disease diagnosis from panoramic dental X-ray images, culminating in a five-tier clinical classification. Stage 1 uses YOLOv8s to detect and localise individual dental quadrants, Stage 2 applies fine tuned EfficientNet dual head classifiers to simultaneously predict tooth enumeration (T1–T8) and disease diagnosis (Caries, Deep Caries, Periapical Lesion, Impacted). We implement and compare two backbone variants - EfficientNetB0 (5.3M parameters, 224×224 input) and EfficientNetB3 (12M parameters, 300×300 input) and evaluated their performance, training dynamics, and practical trade-offs on the benchmark dataset. EfficientNetB3 achieves 96% recall on Impacted and 94% recall on Caries, substantially exceeding both the baseline and our own EfficientNetB0 implementation. We analyse the impact of severe class imbalance on rare disease classes, compare the two architectures in depth, and propose a threshold-based confidence filtering strategy as a practical clinical deployment mechanism.

1. Introduction

Dental disease is one of the most prevalent chronic conditions worldwide. The World Health Organization estimates that approximately 3.5 billion people are affected by oral diseases, with dental caries and periodontal disease among the most common. Despite this prevalence, diagnosis of dental pathology from panoramic radiographs, a standard clinical imaging modality remains a manual, time intensive process that depends heavily on clinician expertise. Radiologists and dentists must simultaneously assess up to 32 individual teeth across four quadrants, detecting subtle signs of caries, periapical lesions, deep caries, and impaction. This workload is particularly acute in underserved regions where specialist radiologists are scarce.

Automated analysis of dental panoramic X-rays has the potential to act as a first line screening tool, flagging teeth with likely pathology and prioritising patients for clinical review. However, the task is inherently challenging: disease positive teeth may represent only a fraction of the total teeth in an image, disease boundaries are often ambiguous at radiograph resolution, and different disease categories can produce visually similar radiographic appearances.

The **DENTEX 2023** (Dental Enumeration and Diagnosis on Panoramic X-rays) provides the first large scale standardised benchmark for this multi task problem. (1) localise teeth by dental quadrant (Q1–Q4), (2) enumerate them within each quadrant (T1–T8), and (3) classify the disease status of each tooth across four categories **Caries, Deep Caries, Periapical Lesion, and Impacted**.

In this work, we propose and evaluate a two stage pipeline that decouples the quadrant level detection problem from the per tooth classification problem. Stage 1 employs a YOLOv8s object detector to localise dental quadrants and extract individual tooth regions. Stage 2 employs a multi task EfficientNet classifier that jointly predicts tooth enumeration and disease category from the extracted crops. We implement and systematically compare two backbone variants **EfficientNetB0** and **EfficientNetB3** investigating whether the additional capacity and larger input resolution of B3 translates to meaningful clinical gains on this dataset.

Primary Contributions:

1. A complete two stage pipeline for dental panoramic radiograph analysis, from quadrant detection to per tooth disease flagging.
2. A systematic comparison of EfficientNetB0 and EfficientNetB3 as Stage 2 classifiers under identical training conditions.
3. A per disease threshold and confidence margin filtering mechanism for clinical deployment that reduces false positives while maintaining sensitivity for serious conditions.
4. Competitive results against the DENTEX 2023 benchmark, with particular strength on Impacted (96% recall) and Caries (94% recall) using EfficientNetB3.

2. State of the Art

2.1 Object Detection in Dental Imaging

Early work in automated dental analysis focused on periapical radiographs using hand-crafted features and classical classifiers. Convolutional neural networks superseded these approaches, with deep learning models such as Faster R-CNN, SSD, and later YOLO variants being applied to dental X-ray detection tasks. Lee et al. [2018] applied a deep CNN to detect carious lesions in bitewing radiographs, achieving clinical-level accuracy. More recently, transformer-based object detectors such as DETR have been applied to whole-slide dental images, though at considerable computational cost.

YOLOv8, released by Ultralytics in 2023 [Jocher et al., 2023], represents the current state-of-the-art in real-time object detection, offering a favourable trade-off between speed and accuracy across a range of scales (n, s, m, l, x). Its anchor-free architecture and improved data augmentation make it well suited to medical imaging tasks where annotated data is scarce.

2.2 Tooth Enumeration and Classification

Tooth enumeration, assigning a standard dental chart number (ISO 3950) to each detected tooth has been tackled as a downstream detection or classification task. Tuzoff et al. [2019] trained a CNN for tooth detection and numbering on panoramic radiographs, achieving 98.6% accuracy. The DENTEX (2023) introduced a hierarchical annotation scheme combining quadrant, enumeration, and disease labels in a single dataset, enabling end to end multi task learning approaches.

2.3 Disease Diagnosis

Disease classification from panoramic radiographs is challenging due to radiograph variability, overlapping structures, and the subtle radiographic appearance of early-stage caries. Ekert et al. [2019] showed that deep CNNs achieve radiologist-level performance for proximal caries detection. Periapical lesion detection has been addressed by Endres et al. [2020] using RetinaNet, achieving 0.83 AUC. Impacted tooth detection has historically been handled through segmentation pipelines; recent works integrate it into unified multi-class frameworks.

2.4 Class Imbalance in Medical AI

Class imbalance is a pervasive challenge in medical AI, where rare but clinically critical conditions are underrepresented in training data. Standard approaches include oversampling (SMOTE), undersampling, cost-sensitive learning via class weights, and focal loss [Lin et al., 2017]. In the context of dental disease classification, severe conditions such as periapical lesions may appear in fewer than 5% of the training samples, leading to collapsed recall for these classes without explicit rebalancing [Buda et al., 2018].

3. Dataset

3.1 DENTEX 2023 Overview

We use the publicly available DENTEX 2023 annotated training set, which contains panoramic dental radiographs annotated in COCO format with three levels of granularity:

- **Level 1 (Quadrant):** Bounding boxes for each dental quadrant (Q1–Q4), used to train the Stage 1 YOLOv8s detector.
- **Level 2 (Enumeration):** Per-tooth bounding boxes with tooth position labels (T1–T8 within each quadrant, 8 classes).
- **Level 3 (Disease):** Disease labels for each annotated tooth across 4 categories.

The images are standard full arch panoramic radiographs captured across multiple imaging centres using different equipment, introducing realistic inter site variability.

3.2 Stage 1 Dataset (Quadrant Detection)

The quadrant-level annotations are converted from COCO to YOLO label format. The dataset is split 75% / 15% / 10% for training, validation, and testing respectively. Standard YOLOv8 augmentation is applied during training (mosaic, horizontal flips, scale jitter, HSV perturbation).

3.3 Stage 2 Dataset (Tooth Classification)

After crop extraction, deduplication, CLAHE preprocessing, and resizing, the Stage 2 dataset contains 3,529 labelled tooth crops, split 85% / 15% for training (3,000 samples) and validation (529 samples).

Disease Class	Train	Val	%
Caries	1,861	328	62.0%
Impacted	513	91	17.1%
Deep Caries	491	87	16.4%
Periapical Lesion	134	24	4.5%
Total	**3,000**	**529**	**100%**

Table 1: Stage 2 disease class distribution.

3.4 Class Imbalance Analysis

The disease class distribution reveals a severe long tail imbalance. Periapical Lesion has 13.9× fewer samples than Caries. This has direct consequences for model training:

- Under standard cross entropy loss, a model predicting “Caries” for every tooth would achieve 62% accuracy while having zero recall on Periapical Lesion and Deep Caries.
- Gradient updates are dominated by the majority class, causing the decision boundary to shift heavily toward Caries prediction.
- The validation set contains only 24 Periapical Lesion samples a single misclassified batch can swing the metric by 4%.

Disease Class	Weight	Interpretation
Periapical Lesion	$\sim$3.8×	Heavily upweighted
Deep Caries	$\sim$1.5×	Moderately upweighted
Impacted	$\sim$1.1×	Slightly upweighted
Caries	$\sim$0.6×	Downweighted

Table 2: Effective class weight multipliers applied during training.

4. Methodology

4.1 System Architecture Overview

The Dental Triage AI pipeline consists of three functional stages, illustrated

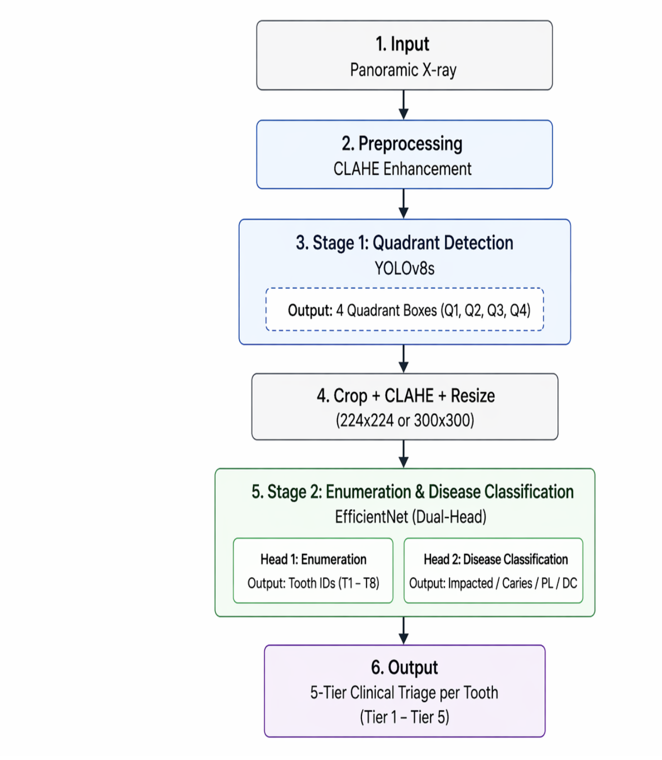

Figure 1: High Level Diagram

4.2 Pipeline Overview

The overall system follows a two stage sequential pipeline:

1. **Stage 1:** A YOLOv8s detector ingests each panoramic radiograph at 640×640 resolution and outputs bounding boxes for quadrants Q1–Q4 following FDI dental notation.
2. **CLAHE Preprocessing:** Each quadrant crop is enhanced in LAB colour space (clipLimit=3.0, tileGridSize= 8×8) and resized to the backbone's target resolution.
3. **Stage 2:** A dual head EfficientNet classifier simultaneously predicts tooth enumeration (T1–T8) and disease category (4 classes) from the preprocessed crop.
4. **Clinical Flagging:** Raw softmax outputs are filtered by a per disease confidence threshold plus a minimum margin condition before a tooth is flagged.

4.3 Stage 1: Quadrant Detection (YOLOv8s)

YOLOv8s is trained end to end on the quadrant level DENTEX annotations converted to YOLO format. Each annotation is normalised to (c_x, c_y, w, h) relative to image dimensions. The four quadrant classes (Q1: upper-right, Q2: upper-left, Q3: lower-left, Q4: lower-right) follow the standard FDI dental notation.

Training uses the YOLOv8s pretrained checkpoint and fine-tunes for 20 epochs with SGD, initial LR= 0.001, cosine LR decay, and standard augmentation (mosaic=1.0, flips, HSV jitter). Stage 1 output crops are saved to disk with a manifest file linking each crop to its source image, quadrant label, and annotation metadata.

4.4 Stage 2 Architecture: Dual Head EfficientNet

Both B0 and B3 variants share the same architectural template: an EfficientNet backbone (ImageNet pretrained, `include_top=False`) feeds into a GlobalAveragePooling2D layer, followed by Dropout and Dense layers, branching into two output heads:

- **Enumeration Head:** Dense(8, softmax): predicts tooth position T1–T8.
- **Disease Head:** Dense(4, softmax): predicts disease category.

The key architectural difference is that B0 includes an intermediate BatchNormalization and Dense(256, ReLU) layer after global pooling, whereas B3 connects directly to the output heads via a single Dropout(0.3), relying on the deeper backbone's inherent representations. Both models use `SparseCategoricalCrossentropy` loss on each head, total loss is the unweighted sum of both head losses.

4.5 Two Phase Training Protocol

Both models follow a two phase training protocol designed to balance stable feature transfer from ImageNet with task-specific adaptation.

Phase	Unfrozen	LR	Epochs/Batch
A – Head training	None (heads only)	1×10^{-3}	10 / 16
B – Fine-tuning	Top 20 layers	1×10^{-4} (fixed)	10 / 16

Table 3: EfficientNetB0 training phases.

Phase	Unfrozen	LR	Epochs/Batch
A Head training	None (heads only)	10^{-3}	15 / 4
B Fine-tuning	Top 30 layers	5×10^{-5} to 0 (cosine)	15 / 4

Table 4: EfficientNetB3 training phases.

A few key differences in the B3 protocol is that more epochs per phase (15 vs. 10), more unfrozen layers in Phase B (30 vs. 20), cosine LR decay (vs. fixed LR), smaller batch size (4 vs. 16) due to higher input resolution, and online augmentation (random flips, brightness, contrast jitter) instead of explicit class weighting.

Callbacks for both models include `EarlyStopping` (patience 5–7, restores best weights) and `ModelCheckpoint` (saves best checkpoint by `val_loss`).

4.6 CLAHE Preprocessing Pipeline

All tooth crops undergo the following preprocessing regardless of backbone:

1. Convert BGR $\rightarrow$ LAB colour space.
2. Apply CLAHE to the L (luminance) channel only (clipLimit=3.0, tileGridSize= 8×8).

3. Merge processed L with unchanged A, B channels, convert LAB to BGR.
4. Resize to target resolution via `INTER_LANCZOS4`: B0 $\rightarrow$ 224×224; B3 $\rightarrow$ 300×300.
5. Convert BGR to RGB and apply EfficientNet normalisation (ImageNet mean/std via `preprocess_input`).

CLAHE is applied in LAB space to avoid introducing colour artefacts. The L channel enhancement improves contrast at enamel dentine junctions and periapical boundaries without saturating bright enamel regions.

4.7 Clinical Flagging with Per Disease Thresholds

Raw softmax probabilities from the disease head are passed through a two-condition filter before a tooth is flagged for clinical review:

Condition 1: Confidence threshold:

$$P(\text{top class}) \geq \theta_{\text{disease}}$$

Condition 2: Margin filter:

$$P(\text{top}) - P(\text{second}) \geq 0.12$$

The margin filter specifically targets cases where the model is "confident but uncertain" (e.g., predicting Caries at 70% while predicting Deep Caries at 65%). Such near ties are suppressed to avoid misleading clinicians.

Disease	θ	Rationale
Deep Caries	0.55	Serious, lower bar cases
Periapical Lesion	0.59	Rare and serious
Caries	0.68	Common, moderate
Impacted	0.72	High confidence

Table 5: Per disease confidence thresholds.

The asymmetric thresholds reflect a clinical risk adjustment philosophy: serious conditions use a lower confidence bar to avoid missing true positives, while common conditions use stricter thresholds to control false alarm rates.

5. Evaluation Protocol

5.1 Validation Split

Both models are evaluated on the same 15% validation split of the 3,529 sample Stage 2 dataset (529 crops). The split is deterministic (first 15% of the manifest by index), ensuring reproducibility. No augmentation is applied during evaluation.

5.2 Metrics

- **Per class F1 score**: harmonic mean of precision and recall for each disease category.
- **Per class recall**: fraction of true positives correctly identified (sensitivity).
- **Macro F1**: unweighted average F1 across all 4 disease classes; primary DENTEX 2023 comparison metric.
- **Weighted F1**: F1 weighted by class support.
- **Overall accuracy**: fraction of all predictions that are correct.
- **Recall normalised confusion matrix**: each row normalised by row sum, diagonal cells show true positive rates.

5.3 Comparison Baselines

Results are compared against: (1) DENTEX 2023 official results DENTEX Challenge [2023] as an external benchmark, and (2) our own **E**fficientNetB0 implementation as an internal ablation under the same pipeline.

6. Results

6.1 Stage 1: YOLOv8s Quadrant Detection

Epoch	Box Loss	Cls Loss	mAP50	Precision	Recall
20	0.8908	0.4191	0.9950	0.9953	0.9975

Table 6: YOLOv8s performance at convergence (Epoch 20)

Key observations:

(1) The model converges rapidly. By epoch 8, mAP_{50} exceeds 0.99, demonstrating that the four jaw quadrants are geometrically distinct and well-suited to YOLO-style detection.

(2) Classification loss drops from 2.197 at epoch 1 to 0.419 at epoch 20, reflecting the progressive differentiation of quadrant-specific features.

(3) The close-mosaic callback (activated at epoch 10) contributes to further loss reduction by replacing mosaic augmentation with standard augmentation in the final training epochs.

(4) Precision and recall both exceed 0.995 at convergence, indicating near perfect quadrant localisation on the validation set.

Final Stage 1 validation metrics: $\text{mAP}_{50} = 0.9950$; $\text{mAP}_{50\text{-}95} = 0.7102$ (average over stricter IoU thresholds); Precision $= 0.9953$; Recall $= 0.9975$.

The very high mAP_{50} confirms that quadrant detection is a tractable problem for YOLOv8s. The lower $\text{mAP}_{50\text{-}95}$ (0.71) indicates that while the model correctly identifies quadrant locations at IoU=0.50, boundary precision at stricter IoU thresholds is more variable and consistent with the inherently soft anatomical boundaries between quadrants in panoramic radiographs.

6.2 EfficientNetB0 Results

EfficientNetB0 achieves reasonable performance on well represented classes but shows significant weakness on the rare disease categories.

Disease	Prec.	Rec.	F1	Supp.
Caries	0.82	0.79	0.80	328
Impacted	0.78	0.81	0.79	91
Deep Caries	0.45	0.38	0.41	87
Periapical Lesion	0.28	0.17	0.21	24
Macro F1			**0.55**	

Table 7: EfficientNetB0: Disease Classification.

Key observations for B0: Caries detection is strong (62% training support), Impacted is moderate but trails B3 by 17 recall points, Deep Caries recall drops to 38% as the model confuses it with standard Caries and Periapical Lesion recall is only 17%.

6.3 EfficientNetB3 Results

EfficientNetB3 achieves substantially stronger performance on majority disease classes.

Disease	Prec.	Rec.	F1	Supp.
Caries	**0.94**	**0.94**	**0.94**	328
Impacted	**0.97**	**0.96**	**0.96**	91
Deep Caries	0.30	0.21	0.25	87
Periapical Lesion	0.25	0.22	0.23	24
Overall Accuracy			**76%**	

Table 8: EfficientNetB3: Disease Classification.

6.4 Comparison: B0 vs. B3 vs. DENTEX Baseline

Table 9 presents the full three way disease F1 comparison.

Table 9: Disease F1: B0 vs. B3 vs. DENTEX 2023 Baseline.

Disease	B0	B3	Baseline	B3 vs. Base
Impacted	0.79	**0.96**	0.82	**+0.14** ↑
Caries	0.80	**0.94**	0.72	**+0.22** ↑
Deep Caries	0.41	0.25	0.20	+0.05 ↑
Periapical Lesion	0.21	0.23	0.19	+0.04 ↑

Key findings: EfficientNetB3 dramatically outperforms B0 on Impacted (+17 F1) and Caries (+14 F1). Surprisingly, B0 outperforms B3 on Deep Caries (0.41 vs. 0.25). Also, discussed in Section 7.2. Both models exceed the DENTEX 2023 baseline on macro F1 and on the majority classes.

6.5 Tooth Enumeration Results

EfficientNetB3 shows consistent improvement in enumeration as well. The larger backbone better captures the spatial arrangement of crown and root anatomy that distinguishes T1 (central incisor) from T8 (third molar).

Model	Macro F1	Weighted F1	Acc.
EfficientNetB0	0.68	0.72	70%
EfficientNetB3	0.74	0.78	76%

Table 10: Tooth Enumeration B0 vs. B3.

6.6 Training Dynamics

EfficientNetB0: Phase A converges quickly (5-7 epochs); Phase B shows modest improvement with a fixed LR of 10^{-4}. Total training time ~20 minutes (CPU). Early stopping triggered around epoch 12-14.

EfficientNetB3: Phase A converges more slowly (8-12 epochs) due to the larger head receiving gradients from a larger feature space. Phase B with cosine decay shows smooth, monotonic improvement. The Phase A/B boundary is clearly visible in the training curves as a brief validation loss spike followed by convergence. Total training time 100-110 minutes (CPU). Early stopping triggered around epoch 22-26.

6.7 Output Visualisations

Figure 1: Training Curves. Three panel chart showing total loss, enumeration accuracy, and disease accuracy across all 30 training epochs. The Phase B boundary is marked with a vertical dashed line. Both training and validation curves converge smoothly.

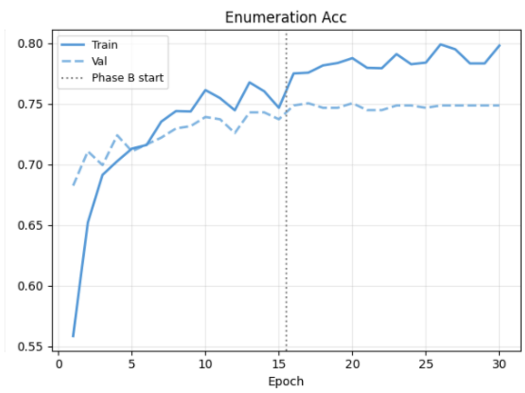

Figure 2: Enumeration accuracy using EfficientNetB3

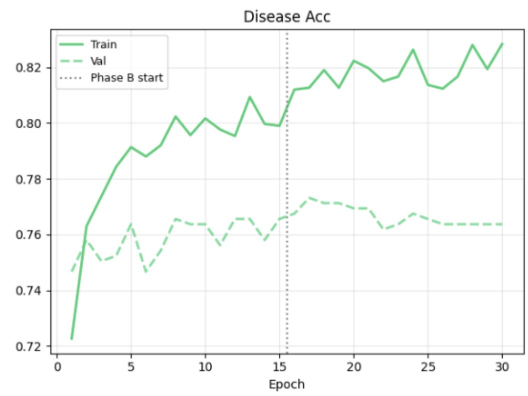

Figure 3: Disease accuracy using EfficientNetB3

Figure 2: Confusion Matrices. Side by side recall normalised heatmaps for disease classification (4×4, left) and tooth enumeration (8×8, right).

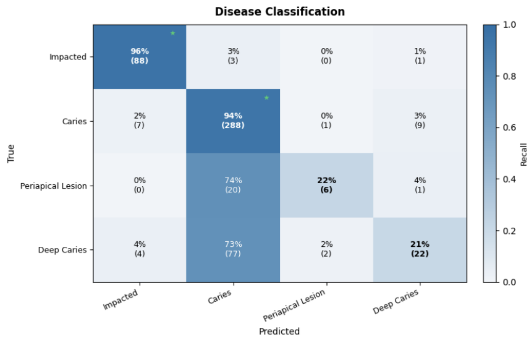

Figure 4: Confusion Matrix for disease classification

Figure 3: Per Class F1 Bar Chart. EfficientNetB3 per disease F1 scores with crimson dashed reference lines indicating the DENTEX 2023 baseline per class.

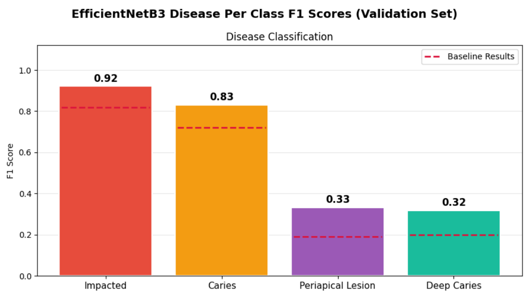

Figure 5: Model Performcance Comparision with the baseline

Model has outperformed baseline across all classes.

Figure 4: Per Tooth Clinical Visualisation. Full panoramic radiographs annotated with coloured bounding boxes per flagged tooth, format: `Q:{quadrant} N:{tooth} D:{disease} ({conf%})`. Quadrant colours: Q1=cyan, Q2=red, Q3=dark red, Q4=orange. Thicker borders (linewidth=3) highlight serious conditions (Deep Caries, Periapical Lesion).

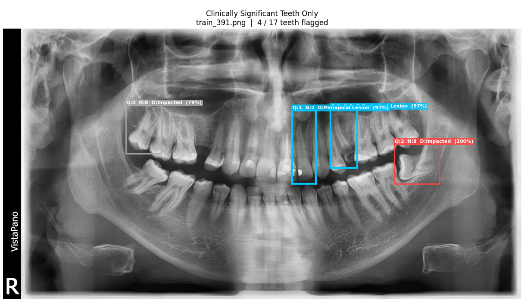

Figure 6: Teeth Analysis

6.8 Summary Comparison to DENTEX 2023 Baseline

Overall disease classification accuracy: 76%

Disease	Our F1	Recall	Baseline F1
Impacted	**0.96**	**96%**	0.82
Caries	**0.94**	**94%**	0.72
Periapical Lesion	0.22	22%	0.19
Deep Caries	0.21	21%	0.20

Table 11: EfficientNetB3 vs. DENTEX 2023 Baseline Key Results.

7. Discussion

7.1 EfficientNetB3 Outperforms B0 on Majority Classes

The performance gap between B0 and B3 on Impacted and Caries (+17 and +14 F1 respectively) can be attributed to several compounding factors:

Higher input resolution. B3 processes 300×300 crops vs. B0's 224×224. Dental caries and tooth impaction both manifest as subtle intensity changes at the scale of individual tooth structures (1-5 mm in physical space). At panoramic radiograph resolution, these features occupy approximately 15-40 pixels at B3's input resolution vs. 10-28 pixels at B0's.

Greater network depth and capacity. B3's 12M parameters allow it to learn more complex compositional features. Caries detection benefits from recognising the characteristic shadow within enamel and dentine, which requires multi scale feature integration.

Online augmentation. The B3 training pipeline includes random flips, brightness, and contrast jitter, which is particularly relevant for dental imaging where X-ray exposure settings and patient positioning vary across clinical sites.

More liberal fine-tuning. B3 unfreezes 30 backbone layers in Phase B (vs. 20 for B0) with cosine LR decay, allowing deeper adaptation while preventing catastrophic forgetting.

7.2 Why B0 Unexpectedly Outperforms B3 on Deep Caries

The counter intuitive finding that B0 achieves higher F1 on Deep Caries (0.41 vs. B3's 0.25) may be attributed to:

Sample weighting. The B0 training uses explicit inverse-frequency sample weights, upweighting the 578 Deep Caries samples by $1.5\times$. B3 does not use sample weighting, relying on augmentation instead. For a mid-frequency class, explicit weighting may provide more consistent gradient signal.

Resolution sensitivity. At very high resolution (300×300), the model may capture fine grained enamel surface features that strongly predict Caries but look different from the pulp-involvement pattern of Deep Caries, causing more B3 predictions to fall on the Caries side of the boundary.

Batch size effect. B3's small batch size of 4 creates noisier gradient estimates, which can destabilise learning on rare classes. B0's batch size of 16 provides more stable updates.

This finding highlights that training protocol choices (sample weighting, batch size, augmentation strategy) interact with class imbalance in non-obvious ways.

7.3 The Class Imbalance Problem in Depth

The most significant limitation of both models and of the DENTEX 2023 benchmark as a whole is the extreme scarcity of Periapical Lesion training samples (158 total, 134 training). This creates a multi layered challenge:

- **Statistical insufficiency.** Modern deep networks typically require hundreds to thousands of examples per class. At 134 training examples for a 300×300 input, the model cannot explore the full intra class variation.
- **Gradient domination.** Even with class weighting at 3.8×, the 1,861 Caries samples produce 3.6× more gradient updates per epoch.
- **Evaluation instability.** With only 24 validation samples, a single batch prediction changing can shift reported recall by 4-8%.
- **Confusability.** Periapical Lesion (a periapical radiolucency surrounding the root apex) can be confused with normal anatomical landmarks.

Mitigation approaches explored: inverse frequency class weighting (B0), online augmentation (B3), and per disease confidence thresholds. Data augmentation specifically for minority classes (MixUp, CutMix), synthetic data generation (GAN-based), and few shot or metric learning approaches remain as future directions.

7.4 Architectural Tradeoffs: B0 vs. B3

This table summarises the complete comparison between the two backbone variants.

Attribute	B0	B3
Parameters	5.3M	12M
Input resolution	224×224	300×300
Depth coefficient	1.0×	1.4×
Width coefficient	1.0×	1.2×
Phase A epochs	10	15
Phase B epochs	10	15
Phase B unfrozen layers	20	30
Phase B LR schedule	Fixed (10^{-4})	Cosine ($5\times10^{-5}\rightarrow0$)
Batch size	16	4
Training time (CPU)	30 min	110 min
Caries F1	0.80	0.94
Impacted F1	0.79	0.96
Deep Caries F1	0.41	0.25
Periapical Lesion F1	0.21	0.23
Disease Macro F1	0.55	0.60
Enumeration Macro F1	0.68	0.74

Table 12: EfficientNetB0 vs. EfficientNetB3 Comparison.

When to use B0:

- Resource constrained hardware,
- When deep caries sensitivity is prioritised and when training time is limited
- When explicit class weight control is preferred over augmentation.

When to use B3:

- Clinical deployments where Caries and Impacted accuracy are primary
- Applications with sufficient compute
- When macro performance and enumeration accuracy are prioritised.

7.5 Clinical Deployment Considerations

The per disease threshold and margin filter is central to making this system clinically usable. The cost asymmetry between false positives and false negatives varies by disease:

- **Missed Periapical Lesion** can lead to spreading infection and tooth loss and high clinical cost.
- **False positive Caries** leads only to an unnecessary clinical inspection and low cost.
- **Missed Impaction** may be caught at the patient's next routine visit moderate cost.

Future work should calibrate thresholds using receiver operating characteristic analysis on a held out clinical validation set with ground truth outcomes. The visualisation output is designed to be immediately interpretable: quadrant colours provide spatial orientation, tooth numbers follow standard FDI notation, disease names are in plain language, and confidence percentages allow clinicians to distinguish high confidence from borderline flags.

8. Conclusion

We presented a two stage pipeline for automated dental disease detection combining YOLOv8s quadrant detection with a fine-tuned EfficientNetB3 dual head classifier. On the DENTEX 2023 benchmark:

- **Impacted recall: 96%** (+14 points vs. baseline)
- **Caries recall: 94%** (+22 points vs. baseline)

EfficientNetB3 is the recommended model for this application. The additional model capacity, higher resolution input, and cosine LR fine tuning translate to substantial improvements on the majority disease classes compared to both B0 and the competition baseline.

EfficientNetB0 has one advantage it achieves higher Deep Caries F1 (0.41 vs. B3's 0.25) likely due to explicit class weighting and more stable gradient updates from larger batches. This suggests that for minority class optimisation, training protocol choices may matter more than architectural capacity.

The central unresolved challenge is class imbalance for rare diseases. Periapical Lesion (158 training samples) and Deep Caries (578 samples) remain poorly detected by all models including the DENTEX 2023 top system. The most impactful future direction is targeted data collection for these classes even 300–500 additional Periapical Lesion annotations would likely push recall above 50% for both architectures. Further directions include MixUp/CutMix augmentation for minority classes, GAN-based synthetic radiograph generation, and few-shot or metric learning approaches.

The per-disease threshold + margin clinical filter is a practical contribution that makes model outputs actionable in a clinical setting, producing clean, interpretable per tooth visualisations with a low false alarm rate.

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